

Topic 5: Arms Race

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Overview



Scenario: Purple (p) and Green (g) are two countries competing in an arms race. Each is:

- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises

Research Question: What happens to the long-term spending of each country?

Model



Arms Race Base Model

Model Equations

$$\frac{\partial p}{\partial t} = ag(t) - bp(t)$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t)$$

- Feasibility Condition: $(p(t), g(t) \geq 0)$
- a, c : Mutual Fear Factor ($a, c > 0$)
- b, d : Expenditure Factor ($b, d > 0$)

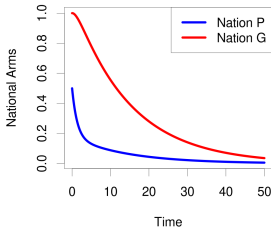


Long Term Behavior (Base Model)

If:

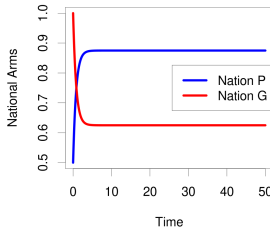
$$bd > ac$$

$$a=0.1, b=0.7, c=0.2, d=0.1$$



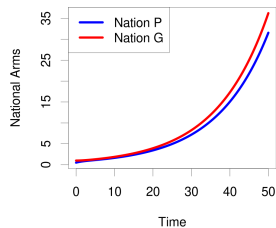
$$bd = ac$$

$$a=0.7, b=0.5, c=0.5, d=0.7$$



$$bd < ac$$

$$a=0.5, b=0.5, c=0.2, d=0.1$$



Complete mutual
disarmament
 $(p^\infty, g^\infty) = (0, 0)$

Stabilized spending
 $(p^\infty, g^\infty) = \left(\frac{cp_0 + bg_0}{a+c}, \frac{dp_0 + ag_0}{a+c} \right)$

Perpetual arms race
 $(p^\infty, g^\infty) = (\infty, \infty)$



Grievances and Goodwill

Suppose countries p and g have underlying goodwill/grievances that influence their spending

The model equations are therefore:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$

- $(r, s) > 0$ for goodwill and $(r, s) < 0$ for grievance
- Provide further elaborating points



Long Term Behavior (Grievances and Goodwill)

- provide plots
- provide stability conditions and convergences
- This model will take on negative values so discuss that by either capping the value or other methods



Mutual Logistic Fear

Limitation of Current Model: Does not account for the limited resources each country has

Solution: Introduce logistic growth

$$\frac{\partial p}{\partial t} = a\left(1 - \frac{p(t)}{K_p}\right)g(t) - bp(t) + r$$

$$\frac{\partial g}{\partial t} = c\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + s$$

- Provide further elaborating points
- Provide further elaborating points

Long Term Behavior (Mutual Logistic Fear)



- provide plots
- provide stability conditions
- provide convergences (if possible)

Thank you for listening!



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